

# Reinforcement-Learning Control of a Cucumber Greenhouse Digital Twin: Results and Assessment

Greenhouses project — control-agent line

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## Abstract

We report the development and honest assessment of a reinforcement-learning (RL) control agent for a calibrated cucumber greenhouse digital twin. Starting from a constant-program baseline (net profit 0.12 EUR/m<sup>2</sup> over held-out autumn weather years), a reactive Soft Actor-Critic (SAC) agent reaches 1.66 EUR/m<sup>2</sup>, and an agent *distilled from a perfect-foresight optimizer* (MPC → RL) reaches 2.83 EUR/m<sup>2</sup> — about 78% of the perfect-foresight ceiling ( $\sim 3.65$  EUR/m<sup>2</sup>). Against the six real growers of the 2018 Autonomous Greenhouse Challenge (AGC), scored on identical economics and a matched crop, the agent is the only net-positive controller at present-day electricity prices, but a price sweep shows this advantage is regime-dependent: the crossover sits near 0.10–0.12 EUR/kWh, and under 2018-era cheap power the best human growers out-earn the agent. The agent *out-economizes* rather than *out-grows*. We conclude the RL line at this point and lay out next steps, foremost among them *price-adaptive agents* that observe and respond to energy and CO<sub>2</sub> prices.

## 1 Context and scope

The Greenhouses project has two goals: (i) a parameter-adjustable greenhouse + crop + photosynthesis model usable as a digital twin, and (ii) an AI agent that controls it. This report covers goal (ii) — the control agent — and its assessment against real data. The twin itself (climate physics, FvCB photosynthesis, and a leaf-cohort cucumber growth model) is documented separately; here we summarize only what bears on control.

## 2 The digital twin (summary)

The twin couples a Vanthoor-type greenhouse climate model (canopy and air temperature, CO<sub>2</sub>, vapour, and a floor thermal mass, solved as a stiff ODE per day) to a Farquhar–von Caemmerer–Berry C3 photosynthesis model and a leaf-age-cohort cucumber growth model with per-organ sink strengths, senescence, and thermal-age fruit cohorts. It was calibrated by t-walk MCMC against all six AGC-2018 growers and reproduces their seasonal yields within about 10% (e.g. Reference grower: model 48.6 vs. observed 49.4 kg/m<sup>2</sup>).

**Twin realism increment (responsive stomata).** We added an opt-in environment-dependent stomatal conductance (Jarvis form, in series with a fixed mesophyll conductance) and a diurnal-corrected maintenance respiration term, both defaulting off so the frozen engine stays bit-for-bit reproducible. A diagnostic established that the autumn scenario is *light-limited*: raising the CO<sub>2</sub> setpoint from 400 to 1200 ppm lifts seasonal gross assimilation by only  $\sim 7\%$  and *reduces* net profit

(dosing cost exceeds the gain). Consequently responsive stomata are a correct-but-second-order refinement for autumn, and CO<sub>2</sub> enrichment has low marginal value — a fact the control results below repeatedly reflect.

### 3 Control problem

**Actions.** The agent sets eleven daily setpoints that a low-level PI/proportional layer turns into the physical actuators: day and night temperature setpoints, CO<sub>2</sub> setpoint, photoperiod (lamp hours), lamp start time, continuous lamp intensity, ventilation proportional band, vent dead-band offset, an outside-temperature screen threshold, a radiation screen threshold, and a vapour-pressure-deficit floor for humidity control. Planting density is an episode-level action.

**Reward.** Daily net profit in EUR/m<sup>2</sup>: fresh-weight revenue (0.889 EUR/kg) minus heating (0.09 EUR/kWh), electricity (0.30 EUR/kWh on-peak, 0.20 off-peak), CO<sub>2</sub> (0.30 EUR/kg), and amortized fixed costs (15 EUR/m<sup>2</sup>/yr). Flue-gas CO<sub>2</sub> from the boiler is credited as free while heating.

**Episodes and splits.** A 100-day autumn crop (plant mid-August) driven by 30 clean historical Dutch weather years, split disjointly into 18 TRAIN, 6 VALIDATION (checkpoint selection), and 6 TEST (never used for training or selection). All figures below are TEST-year net profit at deterministic policy evaluation.

### 4 Controllers

1. **CEM constant program.** A cross-entropy-method search over the best *fixed* season-long setpoints. The “number to beat” and a reward audit.
2. **Reactive SAC.** A hand-rolled Soft Actor–Critic (twin Q-critics, squashed-Gaussian actor, automatic entropy temperature). Naïve training was unstable and lost to CEM; the fix was a warm start behavior-cloned to CEM, a critic-only warmup on a demonstration buffer, a decaying behavior-cloning tether, and demonstration-anchored replay, trained 10<sup>5</sup> steps over 15 seeds.
3. **Perfect-foresight ceiling (MPC).** Because the twin is deterministic given a weather year, the best open-loop schedule with the weather revealed is that year’s optimum — an upper bound on any causal controller. Estimated per TEST year by CEM over a 10-day-block schedule, warm-started from the agent’s trajectory.
4. **MPC → RL distillation.** Forecast-driven expert demonstrations are generated by the open-loop optimizer on each TRAIN year; SAC is then retrained with *state-conditioned* behavior cloning toward those demonstrations, so it imitates a foresightful expert instead of a forecast-blind constant program.

### 5 Results

The distilled agent beats CEM on all six TEST years, turning the two loss-making years strongly positive (1990:  $-1.17 \rightarrow +1.78$ ; 2003:  $-1.50 \rightarrow +1.95$ ) and lifting the mean from 0.12 to 2.83 EUR/m<sup>2</sup>.

| Controller                         | TEST net profit (EUR/m <sup>2</sup> ) | vs. CEM      | seeds beating CEM |
|------------------------------------|---------------------------------------|--------------|-------------------|
| CEM constant program               | 0.12                                  | —            | —                 |
| Reactive SAC (deployed)            | 1.66                                  | +1.54        | 14/15             |
| <b>MPC→RL distilled (deployed)</b> | <b>2.83</b>                           | <b>+2.71</b> | <b>15/15</b>      |
| Perfect-foresight ceiling          | ~ 3.65                                | —            | —                 |

Table 1: The control ladder on held-out autumn weather years. The distilled agent recovers ~ 78% of the perfect-foresight ceiling (the reactive agent, 45%). Per-seed variability also tightened (std 0.30 vs. 0.58).

**The agent learned seasonal timing, not forecast-following.** Evaluating the agent across forecast accuracy (from perfect to noisy) leaves its profit essentially flat (2.79–2.87 EUR/m<sup>2</sup>): it does not condition on the weather forecast in its observation. Yet it reaches 78% of a *perfect-foresight* ceiling. The reconciliation is that the anticipatory value in this problem lives in structure the agent *can* see without a forecast — day-of-season, crop stage, current climate — and the distillation taught it the climatologically optimal program shape. Short-term (3-day) weather foresight has little marginal value in a light-limited autumn; the residual gap to the ceiling is long-horizon perfect foresight that is unrealizable in practice.

## 6 Assessment against real growers (AGC-2018)

We compared the agent to the six real AGC-2018 cucumber growers, scoring both on *our* economics and running the agent on a *matched* crop (the Reference grower’s calibrated parameters, density 2.5) to isolate control quality. The season timing is comparable (~116 real days vs. our 100), but the real weather (Bleiswijk 2018) differs from our historical years, and running the agent on a higher-yield crop than it trained on is out-of-distribution — so results are indicative.

|                             | yield<br>(kg/m <sup>2</sup> ) | heat<br>(kWh) | elec<br>(kWh) | CO <sub>2</sub><br>(kg) | net profit<br>@ 0.25 EUR/kWh |
|-----------------------------|-------------------------------|---------------|---------------|-------------------------|------------------------------|
| AiCU                        | 37.3                          | 116.8         | 118.3         | 9.9                     | −15.19                       |
| Croperators                 | 48.0                          | 233.9         | 183.3         | 14.2                    | −33.75                       |
| DeepGreens                  | 35.8                          | 486.9         | 157.0         | 16.7                    | −61.59                       |
| Reference                   | 49.4                          | 158.2         | 149.3         | 9.8                     | −15.88                       |
| Sonoma                      | 51.3                          | 127.9         | 184.2         | 10.3                    | −20.41                       |
| iGrow                       | 46.8                          | 137.3         | 145.9         | 9.2                     | −15.33                       |
| <b>Agent (matched crop)</b> | 28.3                          | 84.7          | 29.5          | 5.0                     | <b>+4.03</b>                 |
| Agent (native config)       | 22.8                          | 63.2          | 31.6          | 1.2                     | +1.64                        |

Table 2: Real growers vs. agent, common economics. At present-day electricity prices the agent is the only net-positive controller — but see the price sweep.

**The advantage is a price regime, not better growing.** Because the agent’s trajectory is fixed, we swept the electricity price and re-scored everyone (Table 3). The agent’s profit is nearly flat across price (it barely uses electricity); the growers’ profit is steeply price-sensitive (they lit heavily). The crossover is near 0.10–0.12 EUR/kWh. Under 2018-era cheap power (~ 0.05 EUR/kWh) the

best growers — Sonoma 16.4, Reference 14.0, iGrow 13.9 — *out-earn* the agent (9.9). The agent therefore does not grow better; it is price-robust and wins only where power is expensive, by correctly declining supplemental lighting that does not pay in a light-limited autumn.

| net profit (EUR/m <sup>2</sup> )<br>electricity (EUR/kWh) | 0.05   | 0.10   | 0.15   | 0.20   | 0.25   | 0.30   |
|-----------------------------------------------------------|--------|--------|--------|--------|--------|--------|
| AiCU                                                      | 8.47   | 2.55   | −3.36  | −9.27  | −15.19 | −21.10 |
| Croperators                                               | 2.92   | −6.25  | −15.41 | −24.58 | −33.75 | −42.91 |
| DeepGreens                                                | −30.19 | −38.04 | −45.89 | −53.74 | −61.59 | −69.44 |
| Reference                                                 | 13.98  | 6.52   | −0.95  | −8.42  | −15.88 | −23.35 |
| Sonoma                                                    | 16.43  | 7.22   | −1.99  | −11.20 | −20.41 | −29.62 |
| iGrow                                                     | 13.85  | 6.55   | −0.74  | −8.04  | −15.33 | −22.63 |
| <b>Agent (matched)</b>                                    | 9.94   | 8.46   | 6.99   | 5.51   | 4.03   | 2.56   |
| Agent (native)                                            | 7.96   | 6.38   | 4.80   | 3.22   | 1.64   | 0.06   |

Table 3: Electricity-price sweep. The agent’s near-flat line crosses the growers’ declining lines around 0.10–0.12 EUR/kWh. Caveat: the agent’s policy was trained at high prices, so at cheap power it does not light more — a fair cheap-power comparison would retrain it at that price.

**Out-of-distribution diagnostic.** The matched-crop comparison runs the agent on a crop it never trained on, and a trajectory plot exposes the cost: on the Reference grower’s crop the agent is stable for roughly 80 days and then, in the final weeks, emits incoherent setpoints (a night temperature setpoint jumping *above* the day setpoint), a heating spike, and a temperature excursion, while cumulative yield flattens. Running the same agent on its *native* crop under the identical weather year removes the anomaly entirely — the climate is held smoothly to the end, through the same late-autumn cold spell — confirming the cause is not the twin or the weather but the crop-state observations (cumulative yield in particular) leaving the agent’s training range and driving policy extrapolation. The agent is therefore reliable in-distribution and degrades only when pushed outside it; the matched-crop yield (28 kg/m<sup>2</sup>) consequently *understates* the agent’s control quality on that crop, and a short fine-tune on the matched crop (a To-Do item) would remove the end-of-season instability.

## 7 Limitations

Autumn only (a light-limited regime); a 3-day forecast horizon; the matched-crop comparison runs the agent out-of-distribution; the ceiling assumes perfect, physically unrealizable long-horizon foresight; and the headline profit comparison uses present-day (crisis-era) electricity prices under which the growers’ rational-for-2018 strategy looks unprofitable. None of these undercut the internal ladder (all controllers scored identically); they bound the external claims.

## 8 To-Do and future work

**Price-adaptive agents (priority).** The single clearest lesson from the assessment is that a fixed policy is only optimal for the price regime it was trained on. The natural fix is to make *prices part of the problem*: place the current electricity, natural-gas, and CO<sub>2</sub> prices (and, where available, their short-term forecasts) in the agent’s observation, and train across a *distribution* of price regimes

via domain randomization. A single such policy would dial lighting, heating, and CO<sub>2</sub> dosing to the prices it faces — lighting up under cheap power (to rejoin the top growers) and economizing under expensive power (as the current agent does) — rather than committing to one strategy. Two worthwhile extensions: (i) condition on *day-ahead energy-market* prices so the agent shifts lighting and heating into cheap/off-peak hours (temporal load shifting), and (ii) treat gas and electricity prices jointly, since heating and lighting trade off against each other and against CO<sub>2</sub> cost. This directly resolves the regime-dependence found above and turns a brittle single-price controller into a robust, economically rational one.

### Other near-term items.

- **Fair cheap-power comparison:** retrain the agent at  $\sim 0.06$  EUR/kWh and re-run the AGC comparison; test whether it then lights up and matches or beats the top growers at their own prices.
- **Other seasons:** repeat for spring/summer high-light crops, where CO<sub>2</sub>, supplemental light, and stomatal response regain leverage and the twin’s responsive-stomata increment matters; recalibrate the stomatal model per season.
- **Physics:** add the second (blackout/energy) screen; a longer forecast horizon is low priority for autumn but may matter in high-light seasons.
- **Deployable MPC:** a receding-horizon controller with a realistic degrading forecast, both as a strong baseline and a shippable controller; the open-loop optimizer built here is its engine.

### Beyond the control line (other project goals).

- **Inference and probabilistic forecasting:** t-walk inference of crop and climate parameters from real data, and a probabilistic 3–5 week production forecast with an explicit harvest/stopping-time decision and control assumptions.
- **User-friendly simulator:** a configurable, crop-agnostic simulation and training tool exposing the parameters a grower actually sets (climate computer, pruning, actuator configuration), which can also host the trained control agents.

**Reproducibility.** All results come from the twin plus scripts under version control; we recommend consolidating the codebase into a single repository with a shared **GreenhouseCore** engine and separate **Inference**, **Simulator**, and **Control** layers, replacing the current version-folder layout, and versioning milestones by git tag.